

MERIDIAN, MS, USA

WMO: 722340

Lat: 32.335N

Lon: 88.744W

Elev: 90

StdP: 100.25

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13865

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.4	-3.4	-12.3	1.3	0.1	-9.8	1.7	1.5	9.4	14.0	8.5	12.0	2.1	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	35.7	24.4	34.5	24.5	33.4	24.4	26.6	32.1	26.1	31.5	25.6	30.9	3.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.8	24.6	19.8	28.3	24.1	19.2	27.8	83.1	32.1	80.9	31.4	78.8	31.1	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.5	-9.0	37.6	2.5	2.1	-10.8	39.2	-12.3	40.4	-13.8	41.6	-15.6	43.1		
(5)			-9.7	27.7	2.5	0.8	-11.5	28.3	-13.0	28.8	-14.3	29.3	-16.1	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	8.0	10.3	14.0	18.0	22.3	26.0	27.3	27.2	24.4	18.5	12.7	9.3
	DBStd	8.07	5.78	5.56	5.00	3.97	3.20	2.04	1.62	1.91	3.06	4.53	4.78	5.50
	HDD10.0	299	109	59	20	1	0	0	0	0	0	1	26	83
	HDD18.3	1282	323	232	148	54	8	0	0	0	2	54	177	283
	CDD10.0	3298	47	67	146	241	381	480	537	532	433	266	107	61
	CDD18.3	1239	2	6	16	43	131	230	278	274	186	61	9	4
	CDH23.3	12006	6	32	144	430	1247	2274	2759	2728	1723	588	65	11
CDH26.7	5314	0	2	18	90	460	1077	1363	1357	761	178	8	0	
Wind	WSAvg	2.5	3.0	3.2	3.2	3.0	2.5	2.1	1.9	1.9	2.1	2.2	2.4	2.8
Precipitation	PrecAvg	1433	134	134	159	136	109	101	130	105	88	85	110	144
	PrecMax	2008	335	405	418	427	249	279	296	261	274	270	271	376
	PrecMin	899	31	37	13	23	8	7	27	18	2	0	16	28
	PrecStd	282	70	74	83	86	60	60	72	62	64	67	63	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	26.7	28.5	30.4	33.6	36.3	36.8	37.7	36.2	32.6	27.9	24.9
		MCWB	18.0	19.8	19.5	21.0	23.1	24.0	24.3	24.2	23.7	22.0	19.7	19.9
	2%	DB	21.8	23.9	26.7	29.0	32.1	34.6	35.3	35.6	34.0	30.4	25.2	22.6
		MCWB	17.6	18.2	18.3	20.3	22.7	24.0	24.9	24.7	23.6	21.7	18.9	18.9
	5%	DB	19.8	21.8	25.0	27.7	30.8	33.3	34.1	34.2	32.6	28.6	23.2	20.7
		MCWB	16.5	16.7	17.5	19.5	22.1	23.9	25.0	24.7	23.3	20.8	17.7	17.6
	10%	DB	17.7	19.8	22.9	26.1	29.5	32.1	33.0	33.0	31.2	26.9	21.4	18.7
		MCWB	14.9	15.5	16.6	18.6	21.4	23.8	24.9	24.8	23.0	19.9	17.2	15.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	21.0	21.6	23.2	25.1	26.5	27.4	27.4	26.0	25.0	22.4	21.3
		MCDB	22.2	24.1	25.0	27.3	30.3	32.0	32.5	33.0	31.0	28.6	24.6	23.3
	2%	WB	18.8	19.7	20.3	22.0	24.1	25.7	26.7	26.6	25.2	23.8	20.9	19.9
		MCDB	20.7	23.0	23.7	26.2	29.0	31.2	32.1	32.1	30.3	27.8	23.1	21.7
	5%	WB	17.3	18.4	19.1	21.2	23.4	25.3	26.1	26.0	24.7	22.7	19.6	18.4
		MCDB	19.1	21.1	22.9	25.2	28.3	30.7	31.6	31.4	29.5	26.3	22.2	20.2
	10%	WB	15.5	16.6	18.0	20.3	22.8	24.7	25.6	25.5	24.1	21.6	18.0	16.6
		MCDB	17.5	19.0	21.7	24.2	27.5	29.9	31.0	30.7	28.7	24.9	20.6	18.3
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.6	13.3	13.5	12.7	11.6	11.2	11.5	12.3	13.6	13.7	11.9
		MCDBR	13.1	14.0	14.7	14.2	13.8	13.2	12.8	13.3	13.8	14.6	14.9	12.7
		MCWBR	9.0	8.2	7.3	6.5	5.6	4.6	4.4	4.4	4.9	6.7	8.7	8.9
	5% WB	MCDBR	10.5	11.6	11.7	11.3	11.3	11.2	11.1	11.0	10.9	10.8	11.4	10.4
		MCWBR	8.4	7.9	6.9	6.1	5.1	4.5	4.5	4.3	4.4	5.8	8.0	8.3
Clear-Sky Solar Irradiance	taub	0.327	0.330	0.362	0.396	0.430	0.467	0.514	0.511	0.435	0.371	0.343	0.325	
	taud	2.516	2.510	2.431	2.333	2.277	2.208	2.103	2.108	2.303	2.456	2.481	2.536	
	Ebn at Noon	895	928	917	894	862	826	785	782	832	869	867	878	
	Edh at Noon	86	95	110	127	135	145	160	157	124	99	88	80	
All-Sky Solar Radiation	RadAvg	2.74	3.28	4.41	5.53	6.08	6.26	5.98	5.58	5.00	4.19	3.12	2.37	
	RadStd	0.24	0.46	0.37	0.45	0.40	0.47	0.33	0.38	0.37	0.54	0.31	0.24	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	+134	+84

Nomenclature: See separate page